
Neural Scaling Laws for Order Flow Generation

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Abstract

1 Industry-scale order-flow foundation models, including TradeFM [Kawawa-
2 Beaudan et al., 2026] and MarS [Li et al., 2025a], are now being trained with
3 token counts comparable to large language models, yet without a well-defined rule
4 for how parameters and data should be jointly scaled. We show that importing
5 language-model data-budget heuristics into order-flow modelling can lead to severe
6 under-training. Fitting an additive Chinchilla-style scaling law to a Mamba-3 width
7 sweep on tokenised order-flow message streams, we recover a markedly different
8 compute-optimal allocation: near $C \approx 10^{20}$ profiled FLOPs, the optimum lies
9 at tens of order messages per parameter – about 30 messages or 780 tokens per
10 parameter. The fitted data exponent, $\beta \approx 1.33$, is nearly five times the Chinchilla
11 value for natural language ($\beta \approx 0.28$), producing a highly asymmetric loss surface
12 in which under-sampling data is penalised sharply. These results suggest that data
13 budgets for order-flow foundation models cannot be inherited from language-model
14 heuristics, even though the compute-optimal path still couples more data with rapid
15 parameter growth.

16 1 Introduction

17 Limit-order-book (LOB) data is one of the most abundant high-resolution sources of sequential data
18 outside natural language. NASDAQ data [Huang and Polak, 2011] alone records hundreds of billions
19 of order messages across the listed US equity universe, and the totals across other exchanges and
20 other asset classes extend the picture further. Once tokenised, these corpora can reach into the trillions
21 of tokens, comparable in scale to the data budgets used to pretrain modern open-weight language
22 models, opening the door to training very large autoregressive models on order-flow streams.

23 Large autoregressive models for this area face the same allocation question that motivated language-
24 model scaling laws: at a fixed training compute budget, how much should be spent on model
25 parameters and how much on additional observed tokens? The answer cannot be imported directly
26 from natural language. LOB data has stronger local syntax constraints, heavy temporal autocorre-
27 lation, long-tail noise, and a hardware cost profile that is not well captured by the textbook $6ND$
28 approximation, especially for Mamba-style state-space kernels and multi-node training.

29 This paper reports a Mamba-3 logged-loss scaling law. The analysis uses ten Mamba-3 SISO logged-
30 loss curves, maps each checkpoint to $C_{ij} = q(N_i)D_{ij}$ using a profiled FLOPs/token curve, and fits
31 the additive Hoffmann-style surface $L_{ij} = E + A/N_i^\alpha + B/D_{ij}^\beta$. The key technical distinction is
32 that D_{ij} is the recorded tokens seen for checkpoint j of run i , not a fixed multiple of step count, so
33 runs with different batch sizes or tokens per step remain correctly placed.

34 Concretely, the logged-loss fit over 1,846 retained checkpoints from these ten Mamba-3 runs (8M–
35 197M parameters) is

$$L(N, D) = 0.506 + 27.84 N^{-0.370} + 2.18 \times 10^{12} D^{-1.333},$$

and this surface underlies every parametric projection reported in the rest of the paper.

We separate the fitted scaling law from the empirical frontier it summarises. A global additive surface is fit to logged training loss, its compute-optimal projection is compared with the best observed checkpoints, and reconstructed IsoFLOP valleys provide a local check on the same allocation trend. The three views agree on the broad direction of the frontier, while exposing where the smooth fit is only an approximation.

The main conclusion is that order-flow foundation models require their own scaling law. The fitted Mamba-3 frontier differs sharply from the standard Chinchilla allocation for natural language: it is not a universal tokens-per-parameter rule, but a compute-profiled allocation for this architecture, tokeniser, corpus, and observed range.

This paper makes four contributions. We introduce a compute-profiled scaling-law analysis for autoregressive full-book order-flow generation; fit the law on ten Mamba-3 SISO runs spanning 8M–197M parameters and 1,846 logged checkpoints; show that the resulting allocation departs sharply from language-model Chinchilla heuristics; and validate the fitted frontier against empirical best-so-far curves and reconstructed IsoFLOP valleys. We also release the code, FLOPs/token profiles, fitting scripts, and trained checkpoints required to reproduce the analysis using licensed LOBSTER data.

2 Background

Limit order book and LOBSTER. A limit order book records every active resting buy and sell order at one equity on an electronic exchange. Orders at each price level are matched in price-then-time priority, with the best bid and best ask defining the bid–offer spread. Under the continuous double-auction protocol used by NASDAQ, the book evolves through a stream of typed message events: a *new limit order* (Type 1) added to the book if it is non-marketable; a *partial cancellation* (Type 2) reducing the size of an existing resting order; a *full deletion* (Type 3) removing an existing resting order; and an *execution* (Type 4) matching a resting order against incoming liquidity. Hidden executions, trading halts, and cross trades are excluded following Nagy et al. [2023]. The LOBSTER reconstruction [Huang and Polak, 2011] of NASDAQ TotalView–ITCH provides a publicly distributed canonical form of this stream at nanosecond resolution per ticker per day, and the eight-equity sub-corpus used in this paper (Section 3.1) contains roughly 3.09×10^{10} such messages over 2022–2025.

From a raw LOBSTER record to a 26-token block. Each raw LOBSTER record carries a typed mixture of categorical fields (event type, side) and numerical fields (price, size, timestamps, optional reference order ID), so an LOB stream cannot be tokenised in the byte-pair sense used for text. We adopt the field-disjoint tokenisation of Nagy et al. [2023], originally an LOBS5 22-token block over a vocabulary of $|V| = 2,112$ IDs, and extend it to 26 tokens per message via two sequential digit-splits that remove the deep-book price truncation of the original scheme; only a negligible fraction of size clipping remains, and it is disclosed in the preprocessing notes. The resulting per-message layout is shown in Appendix Figure 6, with positions grouped into core message fields, inter-arrival time, absolute time, and reference fields used by cancel/modify/execute messages; the full vocabulary structure and the position-wise field decomposition are deferred to Appendix C. Throughout the paper the data axis D is therefore counted in these 26-token units, not in raw messages or optimiser steps.

Scaling-law functional form and the profiled compute axis. Following Hoffmann et al. [2022] we describe the per-token cross-entropy as

$$L(N, D) = E + A N^{-\alpha} + B D^{-\beta},$$

with N the total parameter count including embeddings (the convention reconciled by Pearce and Song [2024]), D the number of tokens seen, E the irreducible loss floor under our tokeniser, context, and model family, and $\alpha, \beta > 0$ the parameter- and data-side exponents. Turning this surface into a compute-optimal allocation requires fixing a compute axis. The textbook closed form $C \approx 6ND$ [Kaplan et al., 2020, Hoffmann et al., 2022] ignores tokeniser overhead, attention-quadratic terms, and state-space kernels; we instead use a *profiled* compute axis $C = q(N) D$, where $q(N)$ is a FLOPs/token profile measured once per model size from GPU profiler counters (NVIDIA dmon/GPM) during training (Appendix G). After this definition, all compute budgets are reported as

87 C in FLOPs. When $q(N)$ is locally well approximated by kN^γ , the compute optimum has closed
88 form $N^*(C) \propto C^{\beta/(\alpha+\gamma\beta)}$ and $D^*(C) \propto C^{\alpha/(\alpha+\gamma\beta)}$ (Proposition 1; full derivation in Appendix F),
89 so the conventional Chinchilla numerics are recovered only in the special case $\gamma = 1$. Because D is
90 recorded as cumulative tokens seen rather than as optimiser steps, runs with different per-step token
91 budgets (different batch sizes, sequence lengths, or accumulation factors) are placed on the same
92 D -axis without rescaling.

93 **Logged loss, validation CE, and IsoFLOP slices.** The fitted loss in this paper is the *logged*
94 *training loss*, interpreted as a Chinchilla-style pretraining loss in the sub-epoch regime. Hoffmann et
95 al. explicitly model final pretraining loss using smoothed training loss, noting that it is an unbiased
96 estimate of test loss when the number of training tokens is smaller than the full corpus size [Hoffmann
97 et al., 2022]. Our sweep is in the same one-pass accounting regime: the largest run consumes fewer
98 encoded token occurrences than the 2022–2025 LOBSTER corpus contains. We therefore treat logged
99 training loss as the relevant same-distribution pretraining/test loss for the scaling fit. Chronological
100 and instrument-held-out cross-entropy are stricter deployment-shift checks, not prerequisites for
101 fitting the same-distribution scaling surface. An *IsoFLOP valley*, finally, is the locus of (N, L)
102 obtained by interpolating loss curves across runs at a fixed $C = q(N)D$; a parabola in $\log L$ over
103 $\log N$ then identifies the valley minimum $N_{\text{iso}}^*(C)$ and gives a slice-level cross-check on the global
104 additive fit [Czech et al., 2026]. The current paper reports the logged-loss fit and IsoFLOP analysis,
105 while keeping held-out CE as a deployment validation target.

106 **Selective state-space models and Mamba-3.** The architectural family used throughout this paper
107 is the selective state-space model line, in which a recurrent hidden state is updated by an input-
108 dependent transition rather than a fixed kernel. Mamba [Gu and Dao, 2023] introduces this selectivity
109 over diagonal SSMs; Mamba-2 [Dao and Gu, 2024] reformulates it through state-space duality so
110 that selective recurrences run as hardware-efficient blockwise matrix operations during training while
111 preserving constant-memory inference; and Mamba-3 [Lahoti et al., 2026], the variant fit in this paper,
112 additionally applies trapezoidal discretisation, rotary position embeddings on the input-dependent
113 B/C projections, and BCNorm. We use the SISO output variant exclusively (Section 3.2); the MIMO
114 variant and a Transformer counterpart are listed only in the FLOPs/MFU profile of Appendix G. The
115 Mamba-3 block is plugged into a dual-stream message/book/fused outer wrapper at depth allocation
116 $(N_{\text{msg}}, N_{\text{book,pre}}, N_{\text{book,post}}, N_{\text{fused}}) = (2, 1, 1, 6)$; the full schematic of the stream merge and shared
117 sequence-layer wrapper is given by Figure 4 in Appendix A.

118 3 Methodology

119 3.1 Training corpus and held-out evaluation

120 We train on 8 NASDAQ large-cap equities from LOBSTER [Huang and Polak, 2011]: GOOG, AAPL,
121 NVDA, AMZN, META, TSLA, MSFT, and AMD over 2022-01-01 to 2025-12-31. This corpus
122 contains approximately 3.09×10^{10} messages and, under the 26-token-per-message encoding adapted
123 from Nagy et al. [2023], approximately 800B (8.03×10^{11}) tokens at a fixed context of 500 messages
124 (13,000 tokens) per sample.

125 The validation protocol is forward-in-time on January 2026 data, covering all eight in-distribution
126 equities and two out-of-distribution equities, MU and NFLX, that are held out by instrument.

127 This held-out stream defines the validation target for the planned scaling law. The fitted E_{val} would
128 estimate the per-token validation cross-entropy floor for the specified setting: NASDAQ LOBSTER
129 order flow, 500-message context, 26-token encoding, and the chosen Mamba-3 model family.

130 3.2 Architectures and width sweep

131 The reproduced scaling analysis uses only Mamba-3 SISO. The available logged-loss curves comprise
132 10 long runs, ranging from 8M to 197M parameters, using the 26-token message encoding and a
133 500-message context. The Mamba-3 configuration matches the state-space backbone of Lahoti et al.
134 [2026] with SISO output, RoPE on the B/C projections with rope-fraction 0.5, expand factor 2, and
135 state dimension 128. We report the total parameter count N on every plot axis.

3.3 Functional form and scaling fit

We fit the additive Chinchilla-style form

$$L(N, D) = E + AN^{-\alpha} + BD^{-\beta}, \quad (1)$$

In this paper, L is logged training loss. Thus, E in this section is the asymptote of the logged-loss scaling fit rather than a held-out entropy estimate. The parameter term $AN^{-\alpha}$ and data term $BD^{-\beta}$ keep the Hoffmann additive form [Hoffmann et al., 2022], and their fitted exponents describe the Mamba-3 logged-loss surface under the profiled compute convention.

Sample and objective. The fit uses all retained Mamba-3 logged checkpoints after filtering out the early high-loss transient with $L > 1.0$, giving 1,846 rows from ten runs. We minimise the log-Huber objective

$$\sum_i \text{Huber}_\delta(\log L_i - \log \hat{L}_i), \quad \delta = 10^{-3},$$

with $\hat{L}_i = E + AN_i^{-\alpha} + BD_i^{-\beta}$, over the box $\alpha, \beta \in [0.1, 2.0]$, $E \in [0, \min_i L_i]$, $A, B \geq 0$. The outer optimisation over (α, β, E) uses grid initialisation followed by L-BFGS-B polishing; A and B are re-estimated inside the nonlinear objective. The fixed design points are (N_i, D_{ij}) , where D_{ij} is tokens seen at logged checkpoint j of run i .

IsoFLOP cross-check. For each populated slice, we fit a parabola $\log L = a + b(\log N - c)^2$ on its points and compare the empirical compute-optimal size $\hat{N}_{\text{iso}}^*(C) = \exp(c)$ against the parametric prediction under the same profiled compute convention. Slice-versus-parametric mismatch is then attributable to functional-form misspecification, valley noise, or profile error rather than to mixing a profiled C -axis with the analytic $C \propto ND$ convention.

3.4 Uncertainty in the reported figures

The uncertainty bands in Figures 1 and 2 are percentile bootstrap intervals, not standard deviations, standard errors of the mean, or symmetric Gaussian error bars. The grey parametric band resamples temporally correlated logged-loss residuals along the observed training curves under a fixed (N_i, D_{ij}) design, while the red IsoFLOP band resamples retained fixed-compute valley rows. These bands quantify sensitivity of the plotted fits to the observed logged-loss data; they do not include seed-to-seed variation, new training runs, chronological or instrument-held-out validation-loss noise, FLOPs-profile uncertainty, or misspecification of the additive Chinchilla form. We therefore interpret disagreement between the global additive surface and local IsoFLOP valleys as model-specification uncertainty rather than compressing it into a single headline combined CI. Appendix B gives the bootstrap procedures.

3.5 Scaling-law variables and empirical fit scope

The sweep starts from logged LOB training curves rather than from a pre-built scaling-law table. Each retained checkpoint supplies a tuple (N_i, D_{ij}, L_{ij}) , and the analysis is only interpretable once the three axes have fixed meanings.

We therefore use total parameter count for N , cumulative encoded tokens for D , profiled compute $C = q(N)D$, and logged training loss for L . These choices put all runs on the same axes while keeping the claim scoped to the logged-loss surface, not to a held-out validation law.

The same accounting choices also determine how uncertainty should be read. The grey band resamples within-run logged-loss residuals, whereas the IsoFLOP band resamples fitted valley rows; neither band is a seed-replication or new-run interval. Appendix E gives the full checklist and explains how these conventions connect to the methodology critiques in Section 5.

4 Estimating the optimal parameter/training tokens allocation

This section reports the loss-optimal allocation between parameter count and training tokens for Mamba-3 order-flow models. Given a profiled compute budget C , we seek the pair (N^*, D^*) that minimises the fitted logged training loss subject to the measured FLOPs/token curve $q(N)$.

180 All experiments in this section use the Mamba-3 SISO variant only. The analysis uses logged training-
 181 loss curves from 10 Mamba-3 runs spanning 8M–197M parameters, all trained with the 26-token
 182 encoding and a 500-message context. Using the profiled FLOPs/token curve $q(N)$ from Appendix G,
 183 every logged checkpoint is placed at

$$C_{ij} = q(N_i)D_{ij},$$

184 where D_{ij} is tokens seen by run i at checkpoint j . The results should therefore be interpreted as
 185 an empirical logged-loss scaling law: it uses logged/training loss rather than strict chronological
 186 or instrument-held-out validation CE, and the IsoFLOP valleys are sampled from existing long-
 187 run checkpoints rather than from a dedicated matched-compute sweep. Large-scale out-of-sample
 188 validation runs at $C_{\text{val}} = 2 \times 10^{22}$ FLOPs are deferred to a future strict-validation rerun rather than
 189 reported here.

190 4.1 Mamba-3 training-loss scaling fit

191 Each logged checkpoint contributes a tuple (N_i, D_{ij}, L_{ij}) , where N_i is the parameter count of run i ,
 192 D_{ij} is the cumulative number of tokens seen by checkpoint j , and L_{ij} is the corresponding per-token
 193 training cross-entropy. We use tokens seen, rather than optimiser steps, as the data axis. After
 194 omitting the early high-loss transient with $L > 1.0$, the fit uses 1,846 checkpoints across ten runs.

195 We fit the additive Hoffmann-style surface

$$L(N, D) = E + A N^{-\alpha} + B D^{-\beta}$$

196 using a log-Huber objective on log-residuals. The resulting logged-loss fit is

$$L_{\text{logged}}(N, D) = 0.506 + 27.84 N^{-0.370} + 2.18 \times 10^{12} D^{-1.333}.$$

197 Combining this surface with the profiled FLOPs/token curve $q(N)$, under the compute constraint
 198 $C = q(N)D$, gives the projected compute-optimal scaling

$$N^*(C) \propto C^{0.849}, \quad D^*(C) \propto C^{0.236}.$$

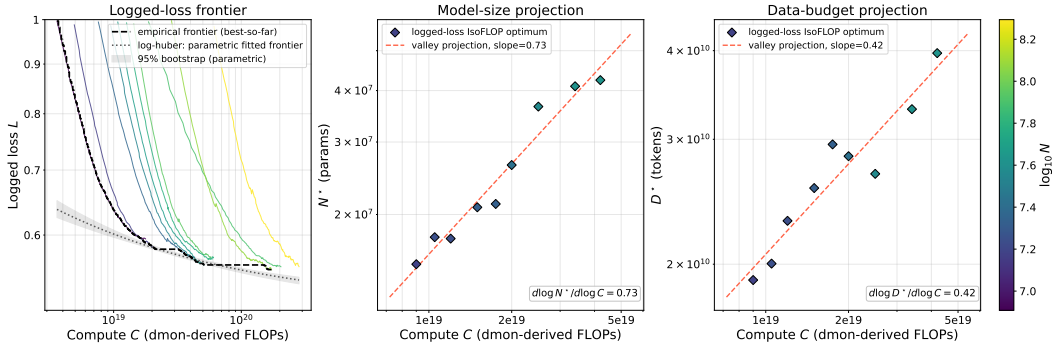


Figure 1: **Mamba-3 logged-loss scaling fit under profiled compute.** *Left:* logged training-loss curves as a function of $C = q(N)D$. The black dashed curve is the empirical best-so-far frontier over observed checkpoints, while the grey dotted curve is the frontier implied by the fitted additive surface under the same compute constraint. *Centre/right:* IsoFLOP valley estimates and their power-law projections for the loss-optimal parameter and data allocations.

199 Figure 1 deliberately separates the observed frontier from the fitted one. The black curve is the best
 200 loss reached by any logged checkpoint up to compute C . The grey curve is the optimum predicted by
 201 the global additive fit under $C = q(N)D$.

202 The difference is evident in the reproduced fit: over roughly 12% of the plotted computational range,
 203 the observed frontier lies below the fitted projection, with the largest gap of about 0.004 nats. We
 204 therefore read the additive law as a smooth global approximation, not as a literal envelope of the runs.
 205 The IsoFLOP valleys provide an independent empirical check of the same allocation trend.

4.2 IsoFLOP valley analysis and uncertainty

As a second check on the allocation trend, we reconstruct fixed-compute slices directly from the logged checkpoints. For each target compute budget C_k , and model size N_i , we compute the corresponding data budget

$$D_{ik} = \frac{C_k}{q(N_i)}.$$

We then select the nearest logged checkpoint for that run and retain it only if its profiled compute is within 2% of C_k . This gives an approximate fixed-compute slice across model sizes.

On each slice, we fit a local quadratic in log model size,

$$\log L = a + b(\log N - c)^2.$$

The fitted valley defines

$$N^*(C_k) = \exp(c), \quad D^*(C_k) = \frac{C_k}{q(N^*(C_k))}.$$

The log-log projection lines in Figure 2 are fitted across the nine recovered valleys. Uncertainty bands are obtained by resampling these nine valley rows 5,000 times and plotting the 2.5–97.5 percentile envelope. These bands measure sensitivity to the recovered valleys only; they do not include new training runs, validation-loss noise, error in the FLOPs/token profile, or misspecification of the additive scaling form. Appendix Table 1 reports the underlying valley estimates and support counts.

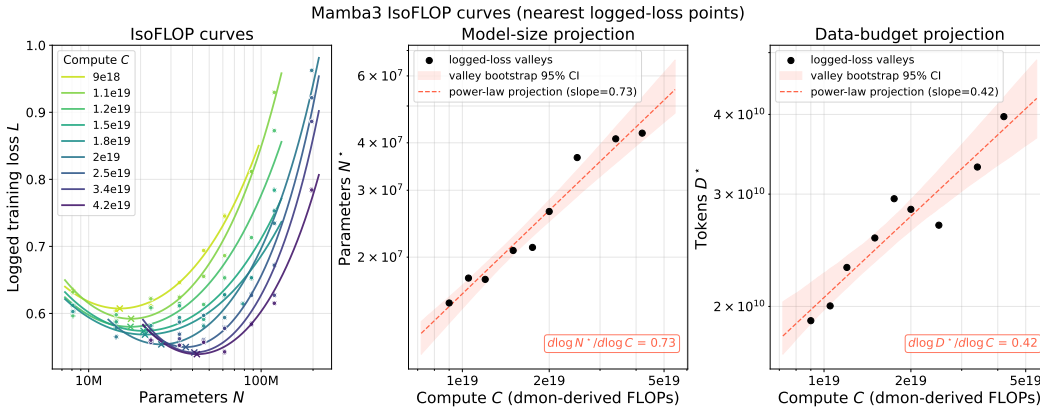


Figure 2: **Logged-loss IsoFLOP valley analysis.** *Left:* approximate fixed-compute slices reconstructed from logged checkpoints, with local parabolas fit in $\log L$ versus $\log N$. *Centre/right:* valley-implied $N^*(C)$ and $D^*(C)$, with log-log projection slopes of approximately 0.73 and 0.42, respectively. Shaded bands are obtained by resampling the nine valley estimates and should be interpreted as valley-resampling uncertainty only.

Each fixed- C slice in Figure 2 has 7–9 model-size points, the left side of several valleys is supported by only two points, and $D^*(C)$ is not strictly monotone. The slopes should therefore be read as local logged-loss allocation trends over the current run set.

4.3 Held-out validation status

The scaling law above follows the Chinchilla convention: it models the same-distribution pretraining loss using smoothed logged training loss in a regime where the consumed training tokens are fewer than the corpus tokens available. Under this one-pass accounting, training loss is the scaling-law loss for the cached pretraining distribution, as in the original compute-optimal language-model analysis [Hoffmann et al., 2022]. The forward-time and instrument-held-out CE pass asks a different question: whether the scaling shape and compute allocation survive the deployment shifts that matter for order-flow modelling.

Those shifts are substantive. First, training examples drawn from the same trading day are highly overlapping, so consecutive logged losses have a smaller effective sample size than the raw token

count suggests. Second, order-book states are strongly autocorrelated within a session, making a uniform shuffle an optimistic deployment protocol; the stricter split is chronological and forward in time. Third, the training corpus covers a finite set of instruments. Structure shared by those instruments can reduce same-distribution pretraining loss while failing to transfer to unseen tickers.

The planned strict-validation pass is designed to measure this deployment gap directly. The saved checkpoints on the same (N_i, D_{ij}) grid will be re-evaluated on a forward-in-time January 2026 stream for the in-distribution tickers, and separately on instrument-held-out tickers such as MU and NFLX. We will then refit the same additive form,

$$L_{\text{val}}(N_i, D_{ij}) = E_{\text{val}} + A_{\text{val}} N_i^{-\alpha_{\text{val}}} + B_{\text{val}} D_{ij}^{-\beta_{\text{val}}},$$

under the same profiled compute convention $C_{ij} = q(N_i)D_{ij}$. The comparison between (α, β) and $(\alpha_{\text{val}}, \beta_{\text{val}})$ will determine how much of the logged-loss allocation trend survives under deployment-relevant validation.

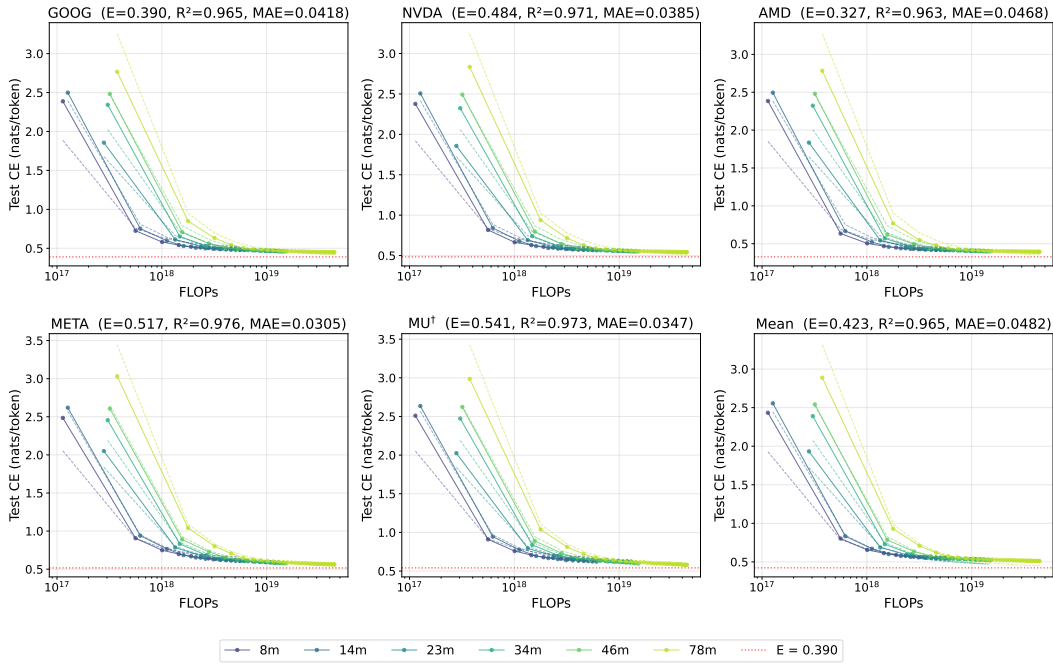


Figure 3: **Per-ticker test-CE scaling trends for Mamba-3 SISO.** The curves use a 500-message context and a January 2026 forward-in-time held-out window. Each panel shows iso-parameter test-CE curves for one evaluation ticker, with α and β locked from an earlier training-loss fit. Instrument-held-out tickers are marked with $\dagger$. The fitted irreducible loss E varies substantially across tickers, from 0.327 on AMD to 0.541 on MU, reflecting differences in message rate and microstructural complexity. This figure documents the previous validation target and motivates the planned strict rerun; it is not part of the current reproduced logged-loss scaling law.

The strict-validation rerun is not a minor numerical refinement. It tests the two shifts that matter for deployment: forward in time and across instruments. An order-flow model is used on future trading sessions and on instruments whose liquidity, volatility, and event-type composition may differ from the training cache. A scaling law that transfers across time but not instruments, or across instruments but not time, is therefore only a description of the cached training population. The logged-loss exponents should be treated as deployment-relevant only if the held-out surfaces reproduce the same allocation trend on both the forward in-distribution stream and the instrument-held-out stream.

4.4 Marginal cost of loss reduction

Under the profiled compute convention $C = q(N)D$, the fitted logged-loss surface implies

$$N^*(C) \propto C^{0.849}, \quad D^*(C) \propto C^{0.236}, \quad L_r(C) = L(C) - E \propto C^{-0.314}.$$

Thus the remaining reducible logged loss falls slowly with compute: halving it would require about $9\times$ more compute, and a decade of additional compute would reduce it by only about a factor of two. This is a local logged-training-loss estimate, not a held-out validation result.

The same frontier also shows why there is no fixed tokens-per-parameter rule. Since

$$D^*(C)/N^*(C) \propto C^{-0.613},$$

the fitted data-per-parameter ratio decreases with compute. Ratios such as “messages per parameter” are therefore local summaries at a chosen compute budget, not constants of the scaling law.

All such statements refer to the fitted parametric surface. They should not be read as claims about the empirical best-so-far frontier, which locally falls below the fitted projection in Figure 1.

5 Related Work

The foundation model paradigm, which has transformed natural language processing, has begun to be adapted for the financial domain. At the aggregated price-and-return level, this includes general time-series foundation models [Ansari et al., 2024, Woo et al., 2024, Das et al., 2024, Liu et al., 2024, Goswami et al., 2024, Garza et al., 2023, Rasul et al., 2024] and the finance-specific Kronos [Shi et al., 2025]. The most granular level of market data, however, is the Limit Order Book (LOB) itself, where the LOBSTER dataset [Huang and Polak, 2011] for the S&P 500 alone amounts to many billions of order messages, which, once tokenised, is comparable in scale to the corpora used to train modern foundation language models such as LLaMA-3 [Grattafiori et al., 2024]. Earlier deep-learning work on LOB data spans CNN-based models [Zhang et al., 2019, Briola et al., 2025], GANs [Coletta et al., 2021, Cont et al., 2023], diffusion models [Backhouse et al., 2025, Berti et al., 2025, Wang and Ventre, 2026], and recurrent networks [Hultin et al., 2023]. Recent work has increasingly focused on large autoregressive foundation models trained on tokenised message sequences, which seem to outperform alternative paradigms on the LOB-Bench distributional benchmark [Nagy et al., 2025]. LOBS5 [Nagy et al., 2023] introduced token-level next-token prediction on tokenized LOBSTER data with an S5 backbone [Smith et al., 2023]; MarketGPT [Wheeler and Varner, 2024] and TradeFM [Kawawa-Beaudan et al., 2026] adopt Transformer backbones, with TradeFM scaling to multi-asset-class trade flow; ByteGen [Li and Chen, 2025] operates directly in byte space for a lossless representation; and MarS [Li et al., 2025a] and LOBERT [Linna et al., 2025] both compress each message into a single token to obtain longer effective contexts, the former within a market simulation engine, and the latter via a BERT-style encoder.

How parameters and data should be scaled jointly is a central question for training such large models. Kaplan et al. [2020] established power-law scaling for autoregressive language models, and Hoffmann et al. [2022] introduced the now-standard additive functional form with an irreducible-loss floor. The same form has since been shown to hold across a wide range of modalities: image, video, and mathematical problem solving [Henighan et al., 2020]; general time-series tokens [Edwards et al., 2024, Yao et al., 2025]; tokenized world-model data [Pearce et al., 2024]; symbolic music tokens [Qu et al., 2024]; motion forecasting and planning [Baniodeh et al., 2025]; and genome-scale nucleotide sequences [Nguyen et al., 2024], with exponents that depend strongly on tokenizer and architecture. Analogous studies report scaling for selective state-space models [Gu and Dao, 2023, Dao and Gu, 2024], gated linear recurrences [De et al., 2024], and the xLSTM family [Beck et al., 2026]. The fitting methodology has correspondingly been sharpened by replication and reconciliation work: Besiroglu et al. [2024] critique overly narrow confidence intervals; Pearce and Song [2024] reconcile Kaplan and Chinchilla by making parameter-counting conventions explicit; Porian et al. [2024] isolate the cosine-LR-decay endpoint and warmup-scaling as dominant sources of the Kaplan–Chinchilla discrepancy; Li et al. [2025b] survey reporting standards across the field; and Czech et al. [2026] characterise structural biases in IsoFLOP parabola fits. We address these critiques in our methodology (Section 3) and report the current Mamba-3 LOB-message scaling law as a reproducible step toward a strict validation scaling law.

6 Conclusion

This paper provides a first scaling-law estimate for tokenised order-flow generation under a measured Mamba-3 compute model. Fitting an additive Chinchilla-style surface to logged training loss, we find that order-flow scaling differs sharply from the language-model heuristic: the fitted data exponent is much larger, and the local compute-optimal allocation corresponds to tens of order-flow messages per parameter rather than a fixed token-per-parameter rule. This suggests that current order-flow foundation models may be undertrained at their reported model scales, and that the optimal allocation depends sensitively on the compute convention used to define the frontier.

The reproduced Mamba-3 sweep gives a coherent logged-loss scaling law, but its allocation frontier is tied to the profiled compute model. With $C = q(N)D$, the fitted optimum scales as

$$N^*(C) \propto C^{0.849}, \quad D^*(C) \propto C^{0.236},$$

rather than following the standard Chinchilla allocation implied by $C \propto ND$. Thus, the main empirical result is not a universal “tokens per parameter” constant, but a profiled allocation frontier for this architecture, tokeniser, corpus, and compute range.

Deployment. This training-time prescription does not require serving the largest trained model in a latency-sensitive trading stack. If inference speed, memory footprint, or co-location cost is the binding deployment constraint, the natural follow-up is to train near the compute-optimal allocation and then distill the resulting autoregressive order-flow policy into a smaller student model, including online policy distillation while the teacher continues to generate or score market continuations. In that use case, the scaling law specifies where to spend training compute, while distillation determines the deployable inference footprint.

Limitations. Several limitations are important for interpreting the scaling figures. First, the reproduced fit is based on the available eight-stock training corpus, corresponding to roughly 800B encoded tokens. We have not yet tested whether the same law holds at larger token budgets, on later forward-in-time streams, or across a broader universe of equities. In particular, the current exponents may partly reflect the liquidity, volatility, event-rate, and tick-size regimes of the stocks used in the sweep. Projections beyond this corpus should therefore be read as extrapolations from the current logged-loss surface, not as validated large-scale or cross-instrument scaling predictions.

Second, the sweep covers only a limited range of model sizes. The current runs span 8M–197M parameters, which is enough to reveal a local allocation trend but not enough to establish a broad asymptotic scaling law over multiple decades of N . This matters especially for the projected compute-optimal frontier, since the fitted exponents are being used to extrapolate beyond the densest region of observed runs.

Third, the longest runs still do not fully populate the large- D regime needed to test the far end of the fitted data exponent. Longer training runs, or a dedicated matched-compute sweep with more evenly spaced checkpoints at high token counts, would be needed to check whether the same D -dependence persists further out.

Finally, the current sweep does not include enough independent seeds at each model size to estimate seed-to-seed or new-run variance. The bootstrap bands quantify within-run logged-loss residual variation and IsoFLOP valley-row sensitivity, but they should not be interpreted as uncertainty over fresh training runs, architecture choices, or new data samples.

Code and checkpoints. Open-source code and trained checkpoints are available at <https://github.com/openreview-submission/anonymous>. The eight-equity LOBSTER message stream used in this paper is licensed for academic research under our LOBSTER agreement and cannot be redistributed; reproducing the scaling analysis therefore requires obtaining LOBSTER access directly from Huang and Polak [2011]. The released repository contains the tokeniser, the FLOPs/token profile, the bootstrap and IsoFLOP fitting code, and the Mamba-3 checkpoints needed to regenerate every figure in the paper from a user-provided LOBSTER snapshot.

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A Model topology diagram

All models reported in this paper share a common outer dual-stream wrapper; only the sequence layer within each block varies. The runs in the scaling-law sweep plug a Mamba-3 SISO block in as the SSM core (Section 3.2) at the default depth allocation $N_{\text{msg}} = 2$, $N_{\text{book,pre}} = N_{\text{book,post}} = 1$, $N_{\text{fused}} = 6$.

Overall model topology. The generative model is a dual-stream encoder–decoder. The *message encoder* embeds L integer tokens via a learned $|V| \times d_{\text{model}}$ table and processes them through N_{msg} SequenceLayers (default 2). The *book encoder* processes the d_{book} -dimensional book-state vector — the depth- L_{book} price-volume snapshot of the order book at each message time — through $N_{\text{book,pre}}=1$ SequenceLayer at native book dimension, projects to d_{model} via a dense layer, and applies $N_{\text{book,post}}=1$ further SequenceLayer at d_{model} ; operating the pre-layer at d_{book} avoids a costly up-projection. The two streams are concatenated, projected from $2d_{\text{model}}$ back to d_{model} , and passed through $N_{\text{fused}}=6$ SequenceLayers in the *fused encoder*; a single output projection maps to next-token logits over $|V|$. The default layer count is $2 + 1 + 1 + 6 = 10$ in the message/fused path, plus the 2 book-side layers, totalling 12. Figure 4 below shows the full schematic, including the message, book, and fused streams and the shared sequence-layer wrapper.

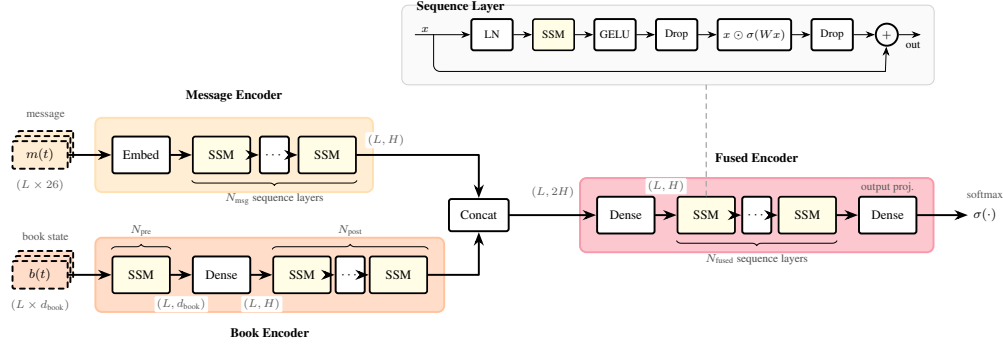


Figure 4: Dual-stream model architecture, adapted from Nagy et al. [2023]. The message encoder (cream) embeds the tokenised message sequence and processes it through N_{msg} sequence layers at dimension $H = d_{\text{model}}$. The book encoder (peach) processes the book-state vector through N_{pre} layers at native dimension, a dense projection to H , and N_{post} layers at H . The two streams are concatenated along the feature dimension and passed through the fused encoder (rose-pink): a projection from $2H$ to H followed by N_{fused} sequence layers and an output projection to vocabulary logits. The yellow-tinted, thick-bordered SSM boxes mark the interchangeable SSM core slots; in this paper every such slot is populated by the same Mamba-3 SISO block (Section 3.2). *Inset*: the shared sequence layer wrapper — LayerNorm, interchangeable SSM core (yellow), GELU, multiplicative gate, dropout, and residual skip. Only the SSM core varies across architecture variants. The module-fill palette (cream / peach / rose-pink) is the same warm ramp as in Figure 6.

B Bootstrap uncertainty procedures

Surface bootstrap. The grey parametric band in Figure 1 is based on a fixed-design residual moving-block bootstrap, not on a label bootstrap over architectures, sizes, or seeds. Let $\hat{L}_{ij} = E + AN_i^{-\alpha} + BD_{ij}^{-\beta}$ be the fitted surface at checkpoint j of run i , and define the log-space residual

$$r_{ij} = \log L_{ij} - \log \hat{L}_{ij}.$$

The design points (N_i, D_{ij}) are held fixed. Within each run, we resample consecutive residual blocks of length 20, concatenate and truncate them to the original run length, and form

$$L_{ij}^{*(b)} = \hat{L}_{ij} \exp(r_{ij}^{*(b)}),$$

then refit Equation 1. The reported surface band uses $B_{\text{boot}} = 200$ refits, with parameter intervals and frontier bands given by the 2.5–97.5 percentiles. Figure 5 gives the visual summary. The band

captures sensitivity to temporally correlated logged-loss noise along the observed curves; it does not estimate seed variance, new-run variance, validation-loss noise, or uncertainty in the FLOPs/token profile $q(N)$.

IsoFLOP projection bootstrap. The red band in Figure 2 uses a separate row bootstrap because its primitive observations are the retained fixed-compute valley rows rather than individual checkpoints. After fitting the nine valleys, each row is

$$(C_k, \hat{N}_k^*, \hat{D}_k^*, \hat{L}_k^*).$$

For each of 5,000 bootstrap draws, we sample nine valley rows with replacement, refit the two log-log projection regressions

$$\log \hat{N}^* = a_N + s_N \log C, \quad \log \hat{D}^* = a_D + s_D \log C.$$

and evaluate them on the plotted compute grid. The red-band envelope is the 2.5–97.5 percentile range of these projection lines. Figure 5 gives the row-bootstrap construction. This is a valley-row sampling band only; it should not be interpreted as a full confidence interval for the scaling law.

Interpretation. We do not assume normally distributed residuals. The surface band captures temporally correlated logged-loss residual variation along the observed training curves under a fixed (N_i, D_{ij}) design. The IsoFLOP band captures sensitivity of the red projection line to resampling the retained valley rows. Neither band includes seed-to-seed variation, new training runs, chronological or instrument-held-out validation-loss noise, FLOPs-profile uncertainty, or misspecification of the additive Chinchilla form; these factors are outside the uncertainty intervals reported here.

Systematic uncertainty. The methodology-critique motivation of Czech et al. [2026] remains relevant: disagreement between the global additive surface and local IsoFLOP valleys is evidence of model-specification uncertainty. In the current paper we expose that disagreement directly in the two figures rather than compressing it into a single headline combined CI. A final validation paper should add a joint prediction band that includes residual bootstrap, seed/new-run variance, FLOPs-profile uncertainty, and surface-versus-IsoFLOP misspecification.

Figure 5 gives the visual summary of the two bootstrap constructions used for the reported uncertainty bands.

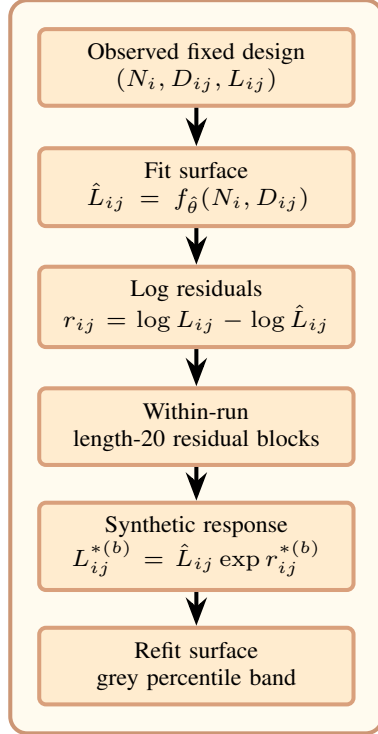
C Tokenisation

This appendix records the technical breakdown of the 26-token encoding introduced in Section 2. Figure 6 gives the raw-record-to-token diagram used by the encoding.

Vocabulary structure. The 2,112-token vocabulary is partitioned by field range, so that token IDs are unambiguous about which field they encode and the per-position syntax masks (below) are trivial to derive. IDs 0–3 are reserved for special tokens (BOS, EOS, PAD, MASK); IDs 4–1003 (1,000 values) form the base-1000 digit pool used by every time field, primary and reference; IDs 1004–1007 encode event type (4 categories: submission, cancellation, deletion, execution); IDs 1008–1107 (100 values) form the base-100 digit pool used by both digits of every size field; IDs 1108–2107 (1,000 values) form the base-1000 digit pool used by both digits of every price-magnitude field; IDs 2108–2109 encode price sign; and IDs 2110–2111 encode buy/sell direction. The total $4 + 1,000 + 4 + 100 + 1,000 + 2 + 2 = 2,112$ matches $|V|$ exactly.

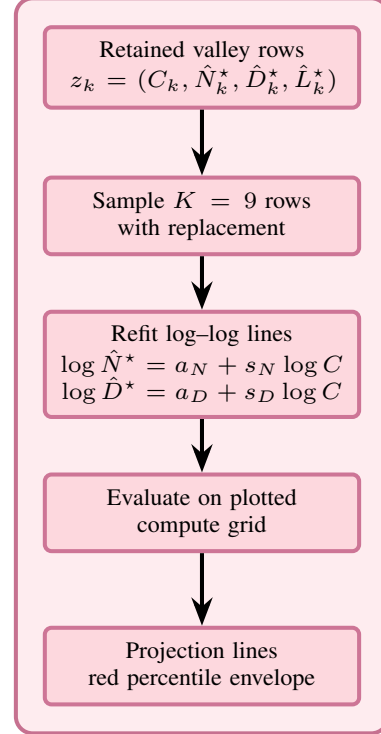
Field encoding. The 26 positions decompose as follows, matching the token labels in Figure 6. Position 0 carries the event type. Position 1 carries direction. Positions 2–4 encode the signed relative price as $(\Delta P \text{ sign}, \Delta P \text{ mag high}, \Delta P \text{ mag low})$, with $p = (-1)^{\Delta P \text{ sign}} \cdot (1,000 \Delta P \text{ mag high} + \Delta P \text{ mag low})$ ticks relative to mid-price, allowing $p \in \pm 999,999$ ticks. Positions 5–6 encode order size as two base-100 digits (Size high, Size low), giving $s = 100 \text{ Size high} + \text{Size low} \in 0\text{--}9,999$ shares. Positions 7–10 encode the inter-arrival time as $(\Delta t_s, \Delta t_{ms}, \Delta t_{\mu s}, \Delta t_{ns})$, where $\Delta t_s \in 0\text{--}999$ is the second component and the sub-second residual is decomposed into three base-1000 digits at millisecond, microsecond, and nanosecond resolution (covering 0 to $10^9 - 1$ ns). Positions 11–15 encode the absolute timestamp as $(t_{ks}, t_s, t_{ms}, t_{\mu s}, t_{ns})$ — five base-1000

Surface residual moving-block bootstrap



conditions on fixed checkpoint design

IsoFLOP row bootstrap



conditions on retained valley rows

Figure 5: Bootstrap procedures used for the reported uncertainty bands. The surface band resamples log residual blocks within each run and refits the additive surface; the IsoFLOP band resamples fixed-compute valley rows and refits the projection regressions. Colours follow the warm palette used in Figure 6.

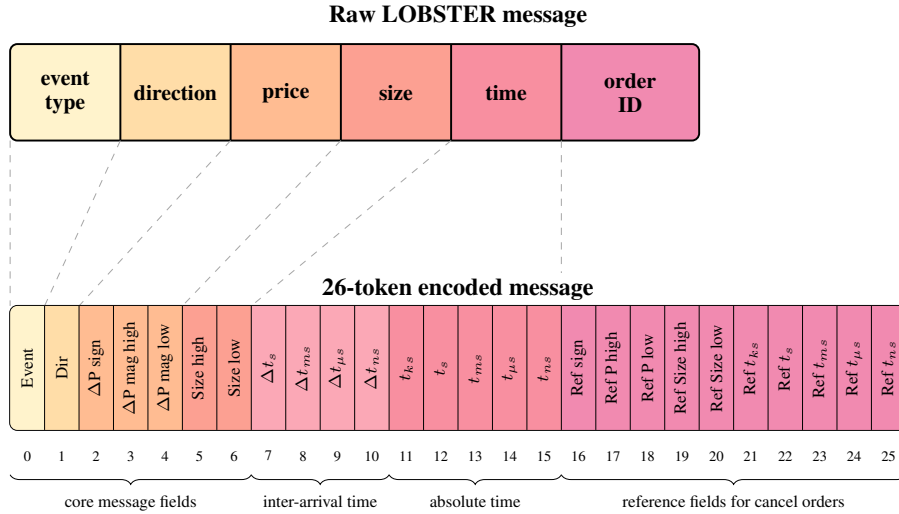


Figure 6: Mapping from a raw LOBSTER message to its 26-token encoding. The top row shows the six fields of a raw record (event type, direction, price, size, time, order ID); the bottom row shows the corresponding fixed-length 26-token block, with dashed connectors linking each raw field to the token positions it expands into. Colours match across the two representations, and the braces below the bottom row group tokens into core message fields, inter-arrival time, absolute time, and reference fields used for cancel orders.

digits at kilosecond, second, millisecond, microsecond, and nanosecond resolution; redundant given Positions 7–10 and the previous message but trained on regardless (see absolute-time paragraph below). Positions 16–25 carry the reference fields used by cancel, modify, and execute messages (LOBSTER types 2, 3, 4): the original order’s signed price (Ref sign, Ref P high, Ref P low), size (Ref Size high, Ref Size low), and absolute timestamp (Ref t_{ks} , Ref t_s , Ref t_{ms} , Ref $t_{\mu s}$, Ref t_{ns}), looked up by order ID at preprocessing time and zeroed for new submissions (type 1). Order ID itself is not tokenised; it serves only as the join key for the reference lookup, and would otherwise be a high-cardinality identifier with no transferable structure across days or tickers.

D Loss decomposition and the FLOP-loss frontier curvature

This section unpacks the formal risk decomposition behind the additive parametric form of Equation 1 and revisits Chinchilla’s own caveat about FLOP-loss frontier curvature [Hoffmann et al., 2022, Appendix E], both of which underpin the methodological choices in Section 3.3 and the validation logic of the planned strict-validation rerun.

The three-term decomposition. Following Hoffmann et al. [2022] (their Equation 9), the population cross-entropy of an empirically-trained predictor $\hat{f}_{N,D}$ admits the orthogonal decomposition

$$L(\hat{f}_{N,D}) = \underbrace{L(f^*)}_{\text{Bayes risk}} + \underbrace{[L(f_N^*) - L(f^*)]}_{\text{functional approximation}} + \underbrace{[L(\hat{f}_{N,D}) - L(f_N^*)]}_{\text{stochastic approximation}}, \quad (2)$$

where f^* is the Bayes-optimal predictor (it outputs the true conditional distribution p of the next token), f_N^* is the population-risk minimiser within the N -parameter hypothesis class $\mathcal{H}_N$, and $\hat{f}_{N,D}$ is the actual stochastic-gradient iterate after one streaming pass over D tokens. The three terms are conceptually orthogonal as flagged in Section 3.3: the first is data-only, the second is capacity-only, and the third absorbs everything that distinguishes a real training run from infinite-data exact minimisation.

Information-theoretic form of the first two terms. Writing each $L(\cdot)$ as a cross-entropy under the next-token conditional distribution p (under the protocol of Section 3.1: 26-token encoding, finite context window, uniformly cycled 8-stock NASDAQ stream) gives

$$L(f^*) = H(p), \quad L(f_N^*) - L(f^*) = \text{KL}(p \parallel q_{\mathcal{H}_N}^*), \quad (3)$$

where $q_{\mathcal{H}_N}^*$ is the distribution induced by f_N^* , equivalently the I-projection of p onto $\mathcal{H}_N$ in KL geometry. The functional-approximation term is therefore a KL divergence between the protocol-restricted next-token distribution and the family, intrinsic to (architecture, tokeniser, context-length) and independent of the optimiser. The Bayes risk $H(p)$ is the conditional entropy under the same protocol, not of the underlying market process; the latter is strictly smaller, and Section 3.1 flags this distinction. In a completed held-out validation law, the constant E_{val} would target this protocol-level Bayes-risk floor, with any residual family-restriction KL not captured by the $AN^{-\alpha}$ term at finite N imparting a small upward bias. The current logged-loss scaling fit reuses the same additive algebra, but its fitted E is a training-loss floor and is not reported as a validated entropy estimate.

Mapping to the additive parametric form. For an ideal held-out validation law, Equation 1 parameterises the three-term decomposition as

$$L_{\text{val}}(N, D) \approx \underbrace{E_{\text{val}}}_{L(f^*)} + \underbrace{A_{\text{val}}N^{-\alpha_{\text{val}}}}_{\text{functional, depends on } N} + \underbrace{B_{\text{val}}D^{-\beta_{\text{val}}}}_{\text{stochastic, depends on } D \text{ only}}. \quad (4)$$

The current reproduced logged-loss fit uses the same separable form with un-subscripted (E, A, α, B, β) as reported fit parameters. The functional power-law $AN^{-\alpha}$ inherits its form from approximation theory [Barron, 1993, Yarotsky, 2017, DeVore et al., 1989], with α set by the smoothness of the target class via covering-dimension arguments. The stochastic power-law $BD^{-\beta}$ inherits its form from the classical one-pass i.i.d. stochastic-approximation rate [Vapnik, 1995, Bottou et al., 2018, Défossez and Bach, 2015], with $\beta \in [\frac{1}{2}, 1]$ depending on margin and noise conditions. Both forms are theoretical templates: in the LOB sweep we fit them as empirical power laws and treat the exponents as free parameters (Section 3.3).

The hidden N -dependence in the third term. Strictly, the stochastic-approximation term $L(\hat{f}_{N,D}) - L(f_N^*)$ depends on *both* N and D : $\hat{f}_{N,D}$ is the SGD iterate inside the N -dimensional landscape, so optimisation difficulty (Hessian conditioning, critical batch size, sensitivity to learning-rate and warmup, occurrence of loss spikes) scales nontrivially with N . The additive separable form $BD^{-\beta}$ in Equation 4 approximates this term as a function of D alone, implicitly assuming that, at fixed setup, the residual N -dependence of $L(\hat{f}_{N,D}) - L(f_N^*)$ is either negligible across the sweep range or empirically absorbed into the fitted constants A, α, B, β . The simplification is not invariant to setup choices: Hoffmann et al. [2022, Appendix G] demonstrate that AdamW with an fp32 optimiser-state weight copy reduces the third term relative to plain Adam at fixed (N, D) , which corresponds to a smaller fitted B rather than a deeper E , confirming that the third term can be sensitive to setup choices the additive form does not directly model. We adopt the same separable form on the LOB grid (Section 3.3) under (i) a single optimiser-and-precision setup across the sweep (Muon kernel + AdamW auxiliary, fp32 master weights, identical LR schedule shape) and (ii) the empirical adequacy of the fit residuals on our (N, D) surface; the assumption is not invariant to optimiser changes that target the μ P or large-batch regimes.

Training-curve geometry of the curvature. A complementary reading recovers the same fixed- C conclusion without further parametric assumptions. Each fixed- N training curve in the $\log C$ - $\log L$ plane bends down toward its own loss floor $E + AN^{-\alpha}$ as $D \rightarrow \infty$, and is tangent to the FLOP-loss frontier at a single compute-optimal point. Whether a single training curve effectively contributes a long or short segment to the frontier depends on the curvature of the frontier near that tangency: when the frontier is locally flat (as a curved-frontier prediction approaches the asymptote) a single curve’s tail tracks the frontier across a wider C window, since the loss penalty for departing from $N^*(C)$ along that window is small; when the frontier is locally steep, the same fixed- N curve falls below the frontier rapidly off-tangency, and a larger N must take over to maintain tangency. A given C range on the frontier is therefore approximated, in a slope-only picture, by a sequence of training curves at increasing N (multiple model-scale-ups, each contributing a short segment), and, in the curved-frontier picture, by fewer training curves at smaller N extended to larger D (each contributing a longer segment). The shift in modal large-language-model practice from successive scale-ups (e.g. GPT-2, GPT-3, MT-NLG) toward extended-token training of fixed-or-shrinking N [Hoffmann et al., 2022, Grattafiori et al., 2024] tracks this geometric distinction; it is the same observation as the Appendix-E remark, restated in the language of training-curve coverage rather than frontier-slope arithmetic.

E Scaling-law variables and empirical fit scope

The object produced by the sweep is a collection of logged loss curves. For run i and checkpoint j , the raw row is

$$(N_i, D_{ij}, L_{ij}),$$

where N_i is the model size, D_{ij} is the cumulative number of encoded tokens processed by that checkpoint, and L_{ij} is the logged per-token training loss. A scaling-law fit is therefore not just a curve fit; it first requires deciding what each axis means.

Parameter axis. We report N as the total parameter count, including embeddings and unembedding. This follows the accounting convention clarified by Pearce and Song [2024]. The choice matters because the fitted surface and the compute-optimal allocation both place model size directly on the horizontal axis.

Data and compute axes. The data coordinate is tokens seen, not optimizer steps. This matters in the present sweep because different runs can have different batch sizes, sequence lengths, or accumulation factors. Counting D as cumulative tokens keeps those checkpoints comparable without an extra step-based rescaling.

The compute coordinate is then built from the profiled FLOPs/token curve,

$$C = q(N)D.$$

Here $q(N)$ is measured once per model size from GPU profiler counters during training (Appendix G). This is the compute convention used in both the parametric frontier and the IsoFLOP slices.

Loss axis. The loss coordinate is logged training loss, specifically the checkpointed `step_loss`. It is not strict chronological or instrument-held-out validation cross-entropy. Logged loss is useful because it is available densely along every run, but it can carry learning-rate-schedule effects and within-run autocorrelation [Porian et al., 2024]. The fitted E is therefore the asymptote of this logged-loss surface, not a validated entropy floor.

Fit and uncertainty. The surface fit uses the additive Hoffmann form with a log-Huber objective on log residuals, with $\delta = 10^{-3}$. This keeps the fit tied to the multiplicative scale on which the residual bootstrap is later defined.

The uncertainty bands are percentile bootstrap bands [Besiroglu et al., 2024], but the bootstrap unit differs by figure. For the grey surface band, the design points (N_i, D_{ij}) are fixed and length-20 residual blocks are resampled within the same run. For the red IsoFLOP projection band, the primitive observations are the nine retained valley rows, which are resampled with replacement.

These intervals should therefore be read narrowly. They capture variation induced by the observed logged-loss residuals or retained valley rows, depending on the figure. They do not include seed variance, new training runs, validation-loss noise, uncertainty in $q(N)$, or full model-specification uncertainty.

Role of the IsoFLOP cross-check. The local IsoFLOP parabolas are used as a diagnostic cross-check rather than as a standalone final estimator. This follows the methodological concern in Czech et al. [2026]: a global additive surface and local fixed-compute valleys need not agree perfectly. We therefore show the disagreement directly, instead of folding it into a single headline confidence interval.

F Profiled compute-optimal allocation

This appendix gives the closed-form derivation of the profiled compute-optimal exponents a, b, p used in Section 4.4. The derivation makes the dependence on the FLOPs/token profile $q(N)$ explicit, so the analytic-compute Chinchilla closed form [Hoffmann et al., 2022, Equation 4] drops out as the special case $\gamma = 1$. All variables are positive throughout.

Proposition 1 (Profiled compute-optimal allocation). *Fix constants $A, B, k > 0$, exponents $\alpha, \beta, \gamma > 0$, and an irreducible loss $E \geq 0$. Assume the additive parametric loss*

$$L(N, D) = E + A N^{-\alpha} + B D^{-\beta}, \quad N, D > 0, \quad (5)$$

and the locally power-law profiled-compute relation

$$C = q(N) D, \quad q(N) = k N^\gamma. \quad (6)$$

For each $C > 0$, the unique minimiser of $L(N, D)$ subject to Equation 6 is

$$N^*(C) = G C^a, \quad D^*(C) = \frac{1}{k} G^{-\gamma} C^b, \quad (7)$$

with closed-form exponents and prefactor

$$a = \frac{\beta}{\alpha + \gamma\beta}, \quad b = \frac{\alpha}{\alpha + \gamma\beta}, \quad G = \left(\frac{\alpha A}{\gamma\beta B k^\beta} \right)^{1/(\alpha + \gamma\beta)}. \quad (8)$$

The compute-optimal reducible loss therefore decays as

$$L_r^*(C) := L(N^*(C), D^*(C)) - E = \left(1 + \frac{\alpha}{\gamma\beta} \right) A G^{-\alpha} C^{-p}, \quad p = \frac{\alpha\beta}{\alpha + \gamma\beta}. \quad (9)$$

Recovery of the analytic Chinchilla case. Setting $\gamma = 1$ and $k = 6$ (the textbook $C \approx 6ND$ assumption) reduces a to $\beta/(\alpha + \beta)$, b to $\alpha/(\alpha + \beta)$, and p to $\alpha\beta/(\alpha + \beta)$, matching Hoffmann et al. [2022, Equation 4] after absorbing the constant 6 into G .

709 *Proof of Proposition 1.* Use Equation 6 to write $D = C/(kN^\gamma)$ and reduce the constrained problem
 710 to a single-variable minimisation in N . Set $\tilde{L}(N; C) := L(N, C/(kN^\gamma)) - E$. Then

$$\begin{aligned}\tilde{L}(N; C) &= A N^{-\alpha} + B \left(\frac{C}{k N^\gamma} \right)^{-\beta} \\ &= A N^{-\alpha} + B k^\beta C^{-\beta} N^{\gamma\beta}.\end{aligned}\quad (10)$$

711 The pink factor $N^{\gamma\beta}$ is the only place γ enters the calculation: in the analytic-compute case $\gamma = 1$ it
 712 collapses to N^β and the rest is the textbook Chinchilla derivation. Differentiating Equation 10 in N
 713 at fixed C ,

$$\frac{\partial \tilde{L}}{\partial N} = -\alpha A N^{-\alpha-1} + \gamma\beta B k^\beta C^{-\beta} N^{\gamma\beta-1}.\quad (11)$$

714 Setting Equation 11 to zero and multiplying through by N rearranges to the *profiled balance condition*

$$\begin{aligned}\alpha A N^{-\alpha} &= \gamma\beta B k^\beta C^{-\beta} N^{\gamma\beta} \\ &= \gamma\beta B \left(\frac{C}{k N^\gamma} \right)^{-\beta} \\ &= \underbrace{\gamma\beta B D^{-\beta}}_{\text{data-side, weighted by } \gamma\beta}.\end{aligned}\quad (12)$$

715 The third line undoes the D -substitution and exposes the structural content of the optimum: along
 716 the profiled frontier the parameter-side reducible term $A N^{-\alpha}$ and the data-side reducible term
 717 $B D^{-\beta}$ are tied in the fixed ratio $\gamma\beta : \alpha$ at every C ; the extra factor γ is what is missing from the
 718 analytic-compute case ($\gamma = 1$).

719 To extract $N^*(C)$ in closed form, divide Equation 11 (set to zero) through by $N^{-\alpha-1}$ to isolate
 720 $N^{\alpha+\gamma\beta}$:

$$\begin{aligned}\alpha A &= \gamma\beta B k^\beta C^{-\beta} N^{\alpha+\gamma\beta} \\ \implies N^{\alpha+\gamma\beta} &= \frac{\alpha A}{\gamma\beta B k^\beta} C^\beta.\end{aligned}\quad (13)$$

721 Taking the $(\alpha + \gamma\beta)$ -th root of both sides gives

$$N^*(C) = \left(\frac{\alpha A}{\gamma\beta B k^\beta} \right)^{1/(\alpha+\gamma\beta)} C^{\beta/(\alpha+\gamma\beta)} = G C^a,\quad (14)$$

722 where G is defined in Equation 8 and $a = \beta/(\alpha + \gamma\beta)$.

723 Recovering $D^*(C)$ from the constraint $D = C/(kN^\gamma)$,

$$\begin{aligned}D^*(C) &= \frac{1}{k} (N^*)^{-\gamma} C = \frac{1}{k} G^{-\gamma} C^{1-\gamma a} \\ &= \frac{1}{k} G^{-\gamma} C^{(\alpha+\gamma\beta-\gamma\beta)/(\alpha+\gamma\beta)} = \frac{1}{k} G^{-\gamma} C^{\alpha/(\alpha+\gamma\beta)},\end{aligned}\quad (15)$$

724 which is Equation 7 with $b = \alpha/(\alpha + \gamma\beta)$. The identity $\gamma a + b = 1$ used in passing from line one to
 725 line two is just the constraint Equation 6 in exponent form: $C = q(N^*) D^*$ becomes $C \propto C^{\gamma a} C^b$.

726 For the reducible-loss exponent p , substitute the balance condition Equation 12 into the reducible
 727 loss along the optimum:

$$\begin{aligned}L_r^*(C) &= A (N^*)^{-\alpha} + B (D^*)^{-\beta} \\ &\stackrel{\text{Eq. 12}}{=} A (N^*)^{-\alpha} + \frac{\alpha}{\gamma\beta} A (N^*)^{-\alpha} \\ &= \left(1 + \frac{\alpha}{\gamma\beta} \right) A (N^*)^{-\alpha}.\end{aligned}\quad (16)$$

728 Combining Equation 16 with $N^* \propto C^a$ from Equation 14 yields $L_r^*(C) \propto C^{-\alpha a}$, and $\alpha a =$
 729 $\alpha\beta/(\alpha + \gamma\beta) = p$ is Equation 9.

Uniqueness of the minimiser: Equation 10 diverges as $N \downarrow 0$ and as $N \rightarrow \infty$, so any critical point is a global minimiser. Moreover, after multiplying Equation 11 by the positive factor $N^{\alpha+1}$, the critical-point equation is equivalent to

$$\gamma\beta B k^\beta C^{-\beta} N^{\alpha+\gamma\beta} - \alpha A = 0,$$

whose left-hand side is strictly increasing from $-\alpha A$ to $+\infty$ on $(0, \infty)$. Hence there is exactly one critical point, and it is the unique global minimiser. The recovered D^* is then determined by the constraint Equation 6. $\square$

Specialisation to the Mamba-3 logged-loss fit. Plugging $\alpha = 0.370$, $\beta = 1.333$, $\gamma = 0.90$ from Section 4.4 into Equation 8 gives

$$a = \frac{1.333}{0.370 + 0.90 \cdot 1.333} \approx 0.849, \quad b = \frac{0.370}{0.370 + 0.90 \cdot 1.333} \approx 0.236,$$

and Equation 9 gives $p \approx 0.314$, recovering the numerical profiled exponents $N^*(C) \propto C^{0.849}$, $D^*(C) \propto C^{0.236}$, $L_r^*(C) \propto C^{-0.314}$ reported there.

G Training protocol

This appendix consolidates the training-protocol details (shared schedule, two-track Muon-AdamW optimiser, compute-axis measurement, and IsoFLOP-slice construction) that the scaling-law analysis of Section 3.3 treats as fixed inputs. The current analysis assumes these optimiser choices transfer across the width sweep; it has not verified an optimiser-invariance window with learning-rate ablations.

Shared schedule and token budget. The reproduced runs use global batch size 128 sequences, weight decay 5×10^{-3} , and a 1%-warmup cosine learning-rate schedule whose denominator coincides with each run’s target step count. The fit itself uses recorded tokens seen, D_{ij} , so differences in effective tokens per step are handled through the data column rather than through a fixed step multiplier.

Two-track Muon-AdamW optimiser. 2D weight matrices are trained with Muon [Liu et al., 2025, Jordan et al., 2024] at a fixed $\eta_{\text{Muon}} = 10^{-2}$ across the entire width grid; embeddings, the unembedding/decoder, layer-norm gains, and biases (the AdamW auxiliary path, $\leq 5\%$ of total parameters and shrinking with d) use a flat $\eta_{\text{AdamW}} = 5 \times 10^{-4}$.

Width invariance of Muon. Muon orthogonalizes the gradient by Newton-Schulz iteration and rescales by $\sqrt{\max(\text{fan-in}, \text{fan-out})}$, producing a per-step activation update bounded independently of matrix dimensions at fixed depth. A single fixed Muon kernel learning rate therefore matches optimization dynamics across the whole width grid without μP -style scaling on the kernel path [Liu et al., 2025].

AdamW auxiliary path and μP fallback. On the AdamW auxiliary path the standard μP prescription [Dey et al., 2024] (input-embedding LR constant, decoder LR $\propto 1/d_{\text{model}}$ if untied, layer-norm and biases constant, AdamW $\epsilon \propto 1/\sqrt{d_{\text{model}}}$) is documented as the conditional fallback to be activated if the flat setting falls outside the invariance window at any width.

LR ablation at $d_{\text{model}} = 1024$. A future validation run should include or cite a width-local LR ablation before treating the fitted loss surface as optimizer-invariant. The current analysis keeps optimizer settings fixed and does not use LR retuning to improve the plotted frontier.

Compute axis. The profiled compute axis defined in Section 2 converts steady-state GPU throughput into a FLOPs/token profile $q(N)$. We do not use the textbook $6N$ approximation as the plotted compute axis because for Mamba-3 the SSM scan, conv1d, gating, rope-fraction position encoding, optimizer overhead, and multi-node communication all contribute real GPU work that $6N$ can miss. The long-run checkpoints are placed at $C_{ij} = q(N_i)D_{ij}$.

772 **IsoFLOP slices.** The current IsoFLOP slices are constructed from existing logged check-
773 points rather than from dedicated matched-compute runs. For each target $C_k \in$
774 $\{9.0, 10.5, 12, 15, 17.5, 20, 25, 34, 42\} \times 10^{18}$ FLOPs and each model size N_i , we compute $D_{ik} =$
775 $C_k/q(N_i)$ and select the nearest logged checkpoint within the relative-compute tolerance. Each
776 retained slice has 7–9 model-size points, with at least two points on each side of the fitted parabola
777 minimum. This procedure characterises the current loss curves; the final validation law should replace
778 it with validation-loss slices or dedicated fixed-compute runs whose valley minima are well bracketed.

Table 1: **Fixed-compute valley points used for the IsoFLOP projection.** Each row is one IsoFLOP slice from existing logged checkpoints. The “left/right” column reports how many selected model-size points fall below and above the fitted valley in $\log N$; all slices are therefore weakly bracketed on the small-model side.

C_k (10^{18} FLOPs)	points	left/right	N^* (M params)	D^* (B tokens)	L^*
9.0	7	2/5	15.1	19.0	0.607
10.5	8	2/6	17.6	20.0	0.592
12.0	8	2/6	17.5	23.0	0.580
15.0	8	2/6	20.8	25.6	0.574
17.5	9	2/7	21.2	29.5	0.568
20.0	8	2/6	26.4	28.4	0.554
25.0	7	2/5	36.6	26.8	0.550
34.0	7	2/5	41.0	33.1	0.542
42.0	7	2/5	42.4	39.7	0.539

779 **Per-row profiled FLOPs/token and MFU.** Table 2 reports the per-row FLOPs/token, per-GPU
780 sustained throughput during SM-busy windows, and model FLOPs utilisation (MFU) against the
781 H100 SXM 989 TFLOP/s BF16 peak, separately for the Mamba-3 SISO and Transformer sweeps
782 at our production topology (gBSZ = 128, 26-token encoding, msg_seq_len = 500). The reported
783 FLOPs/token grows super-linearly with d_{model} on both architectures because the AllReduce overhead,
784 Muon Newton-Schulz step, and (for Mamba-3) the SSM scan and conv1d kernels contribute compute
785 that the textbook $6N$ analytic omits, with the gap widening as node count grows. MFU rises with scale
786 on both backbones — from $\approx 8\%$ at the smallest Mamba-3 row to $\approx 27\%$ at 350M, and from $\approx 17\%$
787 to $\approx 47\%$ on the Transformer side — reflecting the larger arithmetic-intensity-to-bandwidth ratio
788 of dense matmuls at wider d_{model} . Cluster-total throughput during SM-busy windows is recovered
789 as per-GPU TFLOP/s $\times$ nodes $\times 4$; e.g. Mamba-3 350M at 32 nodes sustains $268 \times 128 = 34.3$
790 PFLOP/s, and Transformer 200M at 8 nodes sustains $463 \times 32 = 14.8$ PFLOP/s.

Table 2: **Per-row profiled FLOPs/token, per-GPU sustained throughput during SM-busy windows, and MFU versus the H100 SXM 989 TFLOP/s BF16 peak.** Production topology, gBSZ = 128, 26-token encoding, msg_seq_len = 500.

<i>Mamba-3 SISO</i>						
Size	d_{model}	Nodes	BSZ/GPU	FLOPs/tok	TFLOP/s (per-GPU, busy)	MFU vs 989 peak
0.2M	64	2	16	9.84×10^7	80	8.1%
1M	128	2	16	1.57×10^8	82	8.3%
4M	192	4	8	2.34×10^8	88	8.9%
6M	256	4	8	2.73×10^8	88	8.9%
10M	320	4	8	4.20×10^8	106	10.7%
14M	384	4	8	4.59×10^8	103	10.4%
23M	512	4	8	6.25×10^8	105	10.6%
46M	768	4	8	1.144×10^9	122	12.3%
78M	1024	8	4	1.809×10^9	133	13.5%
120M	1280	16	2	2.891×10^9	168	17.0%
200M	1664	32	1	5.552×10^9	220	22.2%
350M	2048	32	1	8.701×10^9	268	27.1%
<i>Transformer</i>						
Size	d_{model}	Nodes	BSZ/GPU	FLOPs/tok	TFLOP/s (per-GPU, busy)	MFU vs 989 peak
0.2M	64	2	16	1.56×10^8	172	17.4%
1M	128	2	16	2.42×10^8	215	21.7%
4M	192	4	8	3.51×10^8	239	24.1%
6M	256	4	8	4.79×10^8	273	27.6%
10M	320	4	8	6.16×10^8	297	30.1%
14M	384	4	8	7.54×10^8	320	32.4%
23M	512	4	8	1.032×10^9	328	33.1%
46M	768	4	8	1.719×10^9	382	38.6%
78M	1024	8	4	2.528×10^9	422	42.7%
120M	1280	16	2	3.374×10^9	404	40.9%
200M	1664	8	1	4.635×10^9	463	46.9%

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Justification: The conclusion includes a limitations paragraph that states the current scaling law is fit on the available eight-stock, approximately 800B-token corpus and has not yet been validated on a 5T-token held-out or expanded stream. It also states that the current sweep has too few independent seeds per model size to estimate seed-to-seed or new-run variance, so the reported bootstrap bands should not be read as replicated-training confidence intervals.

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Justification: Section 3.1 specifies the LOBSTER training corpus (8 NASDAQ tickers, 2022-01-01 to 2025-12-31, 26-token-per-message encoding, 500-message context) and the forward-in-time January 2026 held-out evaluation, including the two fully held-out OOD tickers (MU, NFLX). Section 3.2 fixes the Mamba-3 SISO architecture (state-space backbone

of Lahoti et al. [2026], RoPE on the B/C projections with rope-fraction 0.5, expand factor 2, state dimension 128) and the 8M–197M parameter sweep. Section 3.3 states the fit objective (log-Huber on log-residuals, $\delta = 10^{-3}$), the parameter box, the optimizer (two-track Muon kernel + AdamW auxiliary), the early-transient $L > 1.0$ filter, and the compute convention $C_{ij} = q(N_i)D_{ij}$ with D_{ij} recorded tokens-seen rather than step count. Section 3.4 documents the residual moving-block bootstrap ($B = 200$ surface, 5,000 IsoFLOP-valley resamples, block length 20). Section 3.5 restates the optimizer/loss-column/parameter-accounting/compute-convention reporting checklist. The full FLOPs/MFU profile $q(N)$ and per-run total parameter counts are reported in Appendix G.

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Justification: Section 3.4 defines the uncertainty bands used for the scaling experiments. The grey surface band is a fixed-design residual moving-block bootstrap over logged-loss residuals within each run: the (N_i, D_{ij}) design is held fixed, residual blocks of length 20 are resampled within each run, and the surface is refit for $B = 200$ bootstrap samples. This captures temporally correlated logged-loss noise along the observed training curves, but not seed variance, new-run variance, validation-loss noise, or uncertainty in the FLOPs/token profile $q(N)$. The IsoFLOP projection band separately bootstraps the nine retained fixed-compute valley rows 5,000 times and refits the log-log projection lines for $N^*(C)$ and $D^*(C)$; it captures valley-row sampling sensitivity only. The plotted bands are 2.5–97.5 percentile bootstrap envelopes, not standard deviations, not standard errors of the mean, and not symmetric Gaussian error bars; the paper states that no Normality assumption is made and that these intervals cover the reported logged-loss scaling analysis rather than a separate held-out validation law.

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1007 the experiments?

1008 Answer: [Yes]

1009 Justification: All Mamba-3 training and FLOPs/MFU profiling runs reported in the paper
1010 were carried out on the Isambard-AI national AI research resource (University of Bris-
1011 tol), using NVIDIA GH200 Grace Hopper Superchip nodes (each pairing a 72-core Arm
1012 Neoverse-V2 Grace CPU with an H100-class Hopper GPU and 96 GB HBM3 + 480 GB
1013 LPDDR5X over NVLink-C2C). Per-run model size, batch size, context length, tokens-
1014 seen budget, and the profiled FLOPs/token curve $q(N)$ used to build the compute axis
1015 $C = q(N)D$ are reported in the methodology section and Appendix G, which together fix
1016 the wall-clock and aggregate GPU-hour cost of every fitted scaling-law point. Bootstrap
1017 and surface-fitting compute (CPU-bound SciPy under Ray with 256 workers and 5 GPUs
1018 registered as Ray resources) was run on the same shared internal Linux node described in
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1047 Justification: The paper focuses on a methodological scaling-law analysis for order-flow
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